

Theodor Vararu

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Software engineer with a 13-year track record of shipping national infrastructure and full-stack web applications. Tech lead on a national NHS service handling 1M+ patient records, with production LLM agents over hybrid search infrastructure. Always tinkering with image fine-tuning, voice cloning, embedding-based search. I love working at the intersection of distributed systems, ML, and developer experience.

EXPERIENCE

National Health Service

Technical Architect (Contract)

Jan 2023 – Apr 2026

- **Manage vaccinations in schools:** Mavis is the national service replacing paper consent forms and recording school-age vaccinations (HPV, flu, MenACWY, Td/IPV 3-in-1). I was the hands-on tech lead from the early Ruby on Rails monolith through to public launch and rollout to 9 immunisation teams, 5,700+ schools, and 1M+ patient records. Built the NHS integrations (PDS, NHS Imms API, CIS2 SSO), observability stack (Splunk via custom Lambda), patient-facing consent journey, and offline-first PWA, across 1,700+ commits as the #2 contributor (I personally sourced the engineer who would take #1 from me). By April 2026, on track to be the most widely used system of its kind in England.
- **Mavis reporting:** Architected and built a Python Flask microservice to expose tabular data and CSV exports for NHS commissioners, with emphasis on materialised views, caching, queues, and PostgreSQL performance optimisation.
- **Gromit:** Built a multi-tenant RAG Slack agent that searches Mavis code and team chat history with source permalinks. Initially in Python and LlamaIndex, ported to Bun single-binary with native ONNX embeddings via bun:ffi, Postgres pgvector + ParadeDB hybrid search, and Claude Agent SDK. The lovable dog is now a core member of the team, helping to draft 19+ Mavis release notes, onboarding new team members, and being spun out as a SaaS product.

Department for Education

Tech Lead (Contract)

Jan 2022 – Dec 2022

- **Find a Lost TRN:** Self-service for 500k teachers in England to recover their TRN, the unique number used to apply for jobs or claim pension. Led 2-6 engineers from hello world to public beta and built the OIDC SSO client now part of Get an Identity. Championed and built the public /performance dashboard. The service processes 50k recoveries per year, with the median user getting their TRN in 38 seconds (before: many days).
- **Teaching Regulation Authority:** Bootstrapped two national services: Apply for Qualified Teacher Status, and Refer Serious Misconduct. Wrote ADRs, interviewed developers, scaffolded the apps, deploy pipeline, CI, and path to live.

Ministry of Justice

Senior Developer (Contract)

Sep 2021 – Dec 2021

- **Manage Supervisions (HMPs):** Brought in for a 2-week accessibility audit and fixes. A few days in, after witnessing my throughput, the delivery manager offered a 3-month retainer to help them deliver on their roadmap.

Meta

Software Engineer

Jan 2021 – Jul 2021

- **Workplace:** Accessibility squad on Meta's enterprise comms product (7M+ paid users incl. Spotify, Starbucks, Nestlé). Triaged and fixed issues toward full WCAG 2.2 compliance; wrote the team's accessibility statement.
- **Empathy Lab:** Designed and built prototypes for an accessibility testing tool, a browser extension that simulates disabilities and common assistive technology interactions.

Department for Education

Tech Lead / Senior Developer (Contract)

Sep 2018 – Nov 2020

- **Apply for Teacher Training:** Led development of the candidate-facing side of the national service handling the UK's entire teacher recruitment pipeline, from hello world to public beta in Nov 2019 (Ruby on Rails, Azure). The 2020 cycle handled over 100k applications. Mentored a team of 5 engineers.
- **Find Postgraduate Teacher Training:** Built features on the public catalogue used to search across 10,000+ courses from 230+ training providers in England (C#, Ruby on Rails). Modernised the DevOps pipeline with Docker, Terraform, and Cypress E2E tests; prototyped Node.js + GraphQL alternatives presented to the architecture group.

Gamesys

Senior Developer (Contract)

Mar 2018 – Sep 2018

- **Bundle performance:** Cut bundle size by 33% on a React/Node SPA serving 260k MAU by removing moment.js and trimming lodash. Wrote performance documentation to guide future work.

Graphit Systems

Senior Developer (Contract)

Aug 2017 – Feb 2018

- **Online shop with 3D configurator:** Sole frontend engineer on an inherited React.js + Three.js codebase with no handover. Got up to speed fast and shipped a redesign, Stripe payments, and legacy browser fixes.

Government Digital Service *Senior Developer (SC Security Clearance)* *Mar 2016 – Aug 2017*

- **Accessible autocomplete:** Built the most widely used open-source combobox component for GOV.UK services. Iterated based on feedback from research with users with disabilities, and an audit by the Digital Accessibility Centre. Gave a talk at Accessibility London. Credited in the GDS blog for defining “Accessibility acceptance criteria”.
- **Government as a Platform:** Worked across the programme on multiple products: a serverless JSON API to generate country data based on the OpenRegister, accessible webchat, data visualisations using D3.js. Managed 2 engineers.
- **GOV.UK:** Rebuilt formats as part of the Migration programme, moving from the legacy publisher to a microservices pipeline. Provided oncall support during second-line shifts, and managed deployments of national infrastructure.

Timecounts *Senior Developer* *May 2015 – Feb 2016*

- **Volunteer management:** Shipped product features (calendar, date picker, CSV import validation) on Node.js + React.js + Rails JSON API. Implemented code review and introduced more rigorous coding and linting standards.

Workangel *Lead Developer* *Sep 2014 – Apr 2015*

- **Employee benefits app:** Led the frontend on two single-page Angular.js applications. Shipped a mobile-first design and modernised the build pipeline with minification, linting, and CI. Kicked off the work on a React.js rewrite.

ReadForward *Lead Developer* *Mar 2013 – Sep 2014*

- **Presidential candidate website:** Built the open-source website for Monica Macovei’s 2014 Romanian presidential campaign. Scored 100/100 on Google PageSpeed Insights via critical-path CSS generation and SPA prerendering.
- **Digital textbooks:** Led an open-source digital textbook authoring tool (Node.js, Backbone.js, visual regression testing). It became the company’s main product and the textbooks were adopted by Eton College and other prestigious schools.
- **George Enescu festival:** Rewrote the festival website end-to-end (Ruby on Rails, scraped legacy PHP backend, full-text search, JSON API for native mobile apps). Scaled to 100k+ unique visitors per month.

EDUCATION

- **London South Bank University** *London, UK*
Bachelor of Engineering, Computer Systems and Networks. Transferred from University of Bucharest *2014 – 2016*

RECENT PROJECTS

- **braindance (2026):** I’m building a spatial computing shell for VITURE XR glasses. It captures desktop screens via a zero-copy GPU pipeline and renders them as head-tracked 3D windows. Multi-threaded Rust + Bevy ECS, FFI bindings to the VITURE SDK, and a bespoke 3DoF-to-6DoF tracking pipeline (OpenCV ChArUco + PnP, IMU fusion).
- **tuicraft (2026):** I built a WoW chat client for the terminal, designed for OpenClaw and other AI agents to play the MMO. Built with Bun, 100% tested, while heavily leveraging AI code generation. Open source, AGPLv3.
- **zp-voicemod (2026):** I reverse engineered the ZERO PARADES game and wrote a pipeline that trains a voice model (hubert-bsqvae-small) on a target voice & patches every dialogue line with your voice. Open source, GPLv3.
- **untitled.ink (2025 – 2026):** I’m building an interactive fiction platform, featuring a Claude Code-style agentic editor, the Ink markup language, hot-reloading, and video/image generation & editing on Replicate or ComfyUI.

SKILLS

- **Languages:** TypeScript, JavaScript, Ruby, Python, Rust, C#
- **Tools:** Bun, React.js, Next.js, Node.js, Astro.js, Flask, Ruby on Rails, Hotwire, Tailwind, PostgreSQL, Redis, SQLite
- **Infrastructure:** AWS, Azure, Docker, Terraform, GitHub Actions, Kamal, DigitalOcean, Vultr, CI, CD
- **Expertise:** Accessibility (WCAG), testing (Playwright, RSpec, Jest, TDD), security, performance, design systems
- **AI / ML:** RAG, ComfyUI, agentic workflows, LlamaIndex, local LLMs (Ollama), embeddings, Replicate, Claude Agent SDK, Claude Code, Codex, OpenCode, MCP, pgvector, voice conversion (VC), OpenClaw, OpenCV